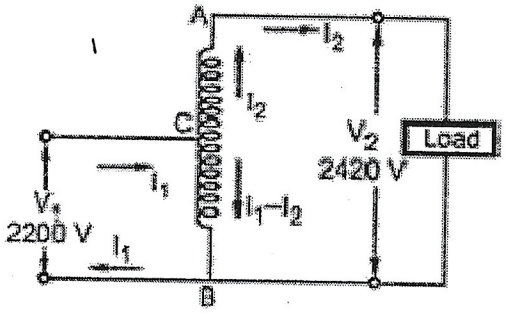
EE 3014/3042 ANSWER SHEET
2024 SPRING FINAL EXAM

EE3042/EE3014 Final Exam Duration: 105 minutes **Date: 11 June, 2024**

Student Name and Number

The notes and books will be kept closed. The calculator can be used. All necessary formulas are provided.

1) (25p) A 2200/220 -V, 20-kVA, two winding transformer is connected as an autotransformer to transform 2200 220V to 2420 V. Find **a)** The transformation ratio of autotransformer, **b)** The kVA rating of autotransformer, and **c)** The currents in various parts of winding. i.e., the currents C through A, C through B, and primary and secondary currents)



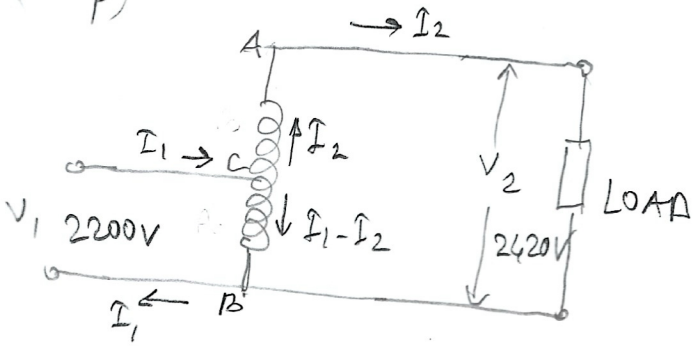
2) (25p) A 230-V, 50 Hz, 2-pole synchronous motor draws 40A from the line at unity power factor and full load. Assuming that motor is lossless, answer the followings: **a)** What is the output torque of this motor?, **b)** What must be done to change the power factor to 0.85 leading? Explain your answer, using phasor diagrams, **c)** What will the magnitude of the line current be if the power factor is adjusted to 0.85 leading?

3) (25p) During a short-circuit test, a Y-connected synchronous generator produces 100 A of short-circuit armature current per phase at a field current of 2.5 A. At the same field current, the open-circuit line voltage is measured to be 440 V. **a)** Calculate the saturated synchronous reactance under these conditions, **b)** If the armature resistance is 0.3 Ω per phase, and the generator supplies 60 A to a purely resistive Y-connected load of 3 Ω per phase at this field current setting, determine the voltage regulation under these load conditions.

4) (25p) A The power crossing the air gap of a 60 Hz, four-pole induction motor is 25 kW, and the power converted from electrical to mechanical form in the motor is 23.2 kW. **a)** What is the slip of the motor at this time? **b)** What is the induced torque in this motor? **c)** Assuming that the mechanical losses are 300 W at this slip, what is the load torque of this motor?

$P_{in} = \sqrt{3}V_T I_L \cos\theta$	$Q = V_s I_s \sin\theta$ per phase	$P = 3V_s I_s \cos\theta$
$Z_{base} = \frac{V_{base}^2}{S_{base}}$	$s = \frac{n_{sync} - n_{rotor}}{n_{sync}}$	$\tau_{ind} = \frac{P_{AG}}{w_{sync}} = \frac{P_{conv}}{w_m}$
$P_{out} = P_{conv} - P_{F\&W} - P_{misc}$ $n_{sync} = 120\frac{f_{se}}{P}$	$P_{conv} = P_{AG} - P_{RCL}$ $VR = \frac{V_{T,nl} - V_{T,f}}{V_{T,f}} x100\%$	$P = \frac{3V_\phi E_A \sin\delta}{X_S}$ $P_{conv} = (1-s)P_{AG}$

1) 2200/220 V, 20 kVA, to transform 2200V to 2420V.
(25p)



Two winding transformer
= 2200/220V
VA rating = 20kVA

The current in the winding CB (2200V) $\Rightarrow I_1 = \frac{\text{VA rating}}{2200} = 9.091\,A$ 3/

The current in the winding CA (220V) $\Rightarrow I_2 = \frac{\text{VA Rating}}{220} = \frac{20\times10^3}{220} = 90.91A$ 3/

a) The transformer ratio (a) $\Rightarrow \frac{V_2}{V_1} = a = \frac{2420}{2200} = 1.1$ 2/
where $K > 1$ active as step up transformer.

b) KVA rating of autotransformer.
kVA rating $= V_2 I_2 = (2420)(90.91) = 220,0022\,kVA$ 5/

c) Current through port AC $\Rightarrow I_2 = 90.91A$ 2/
Current through port BC $\Rightarrow I_1 - I_2 \Rightarrow a = \frac{I_1}{I_2}$
$\Rightarrow I_1 = a I_2 = (1,1)(90.91) = 100,001\,A$ 5/
Current through BC $\Rightarrow$ $I_1 - I_2 = 100.001 - 90.91 = 9.091\,A$ 5/

2) 230V, 50Hz, 2-pole SM
(25p) $I_L = 40A$
PF = 1
No Loss.

a) τ_L = ? b) PF = 1 to PF = 0,85 leading
c) I_L = ? at PF = 0,85 leading

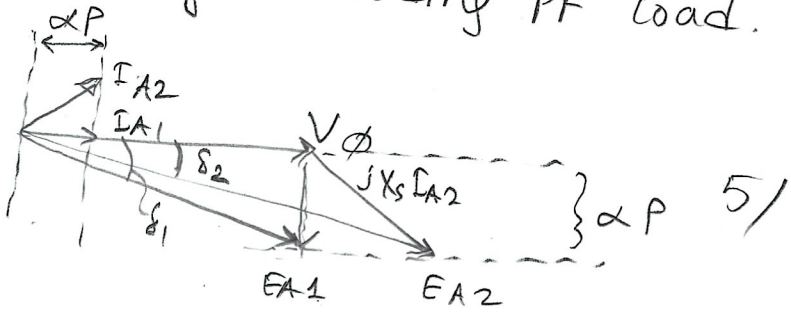
a) $P_{IN} = \sqrt{3} V_T I_L \cos\theta = \sqrt{3}\,(230V)(40A)(1.0) = 15.93kW = P_{out}$ 5/

$n_m = \frac{120 f_{se}}{P} = \frac{(120)(50)}{2} = 1500\,r/min$ 3/

$\tau_L = \frac{P_{out}}{\omega_m} = \frac{15.93\,kW}{(1500\,r/min)\left(\frac{1\,min}{60}\right)\left(\frac{2\pi\,rad}{}\right)} = 101.4\,N,$ 2/4

2) continued

b) If I_F is increased the SM's power factor approaches leading PF, further approach causes to motor running as leading PF load.



5/

3/ Since $I_A \cos\theta$ and V_T is constant, the power supplied to the load is independent of the field current (I_F). $|E_A|$ increases with I_F while keeping the distance $E_A \sin\delta$ constant. The increas in E_A changes the angle of the current I_A, means modifying the PF towards the leading power factor.

c) The magnitude of the line currents will be

$$I_L = \frac{P}{\sqrt{3} V_T PF} = \frac{15.93\,kW}{\sqrt{3}\,(230)(0,85)} = 47\,A \quad 3/$$

3) (25p) Y-connected SG, with short circuited test $\Rightarrow I_A = 100\,A$/phase (short circuited at $I_F = 2.5\,A$)

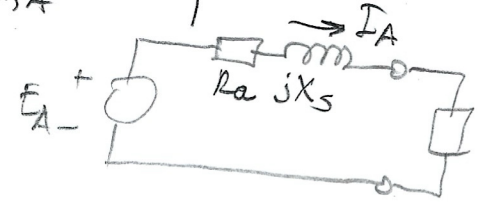
with open circuited test

$V_L = 440\,V$ (open circuited) at $I_F = 2.5A$

a) $X_S = ?$ b) $R_A = 0,3$, $I_A = 60A$ at PF=1, Y-connected load, 3-Ω/phase, $I_F = 2.5A$ } $VR\% = ?$

4) $V_\phi = \frac{V_L}{\sqrt{3}} = \frac{440}{\sqrt{3}} = 254\,V$ 3/

$X_S = \frac{V_\phi}{I_{shrt}} = \frac{254}{100} = 2,54\,\Omega$ 5/



b) With $V_L = 440V \Rightarrow V_\phi = 254\angle 0°$, $I_A = 60\angle 0°$ at unity PF 3/4

$E_A = V_\phi + R_A I_A + jX_S I_A = 254\angle 0° + (0,3 + j2.54)\,60\angle 0° = 312\angle 29.3°$ 8/

3-) continued This is the phase voltage at no load. Because $V_L = 440\,V$ is the open circuit voltage

$$V_{L,nl} = (312)(\sqrt{3}) = 540\,V \quad 4/$$

The voltage regulation is

$$VR = \frac{V_{T,nl} - V_{T,fl}}{V_{T,fl}} = \frac{540-440}{440} \times 100 = 22.7\,\% \quad 5/$$

4-) $P_{AG} = 25\,kW$, $f_{se} = 60\,Hz$, 4-pole, $P_{conv} = 23.2\,kW$

(25p) a) $s = ?$ b) τ_{ind} c) $P_{mech\ losses} = 300\,W$, $\tau_{load} = ?$ at the same slip

a) $n_{sync} = \frac{120 f_{se}}{P} = \frac{120(60)}{4} = 1800\ r/min$ ← synchronous speed of the motor 3/

The converted power from electrical to mechanical form is

$$P_{conv} = (1-s)P_{AG} \Rightarrow s = 1 - \frac{P_{conv}}{P_{AG}} = 1 - \frac{23.4\,kW}{25\,kW}$$

$\Rightarrow s = 0,064$ or $s = 64\,\%$ 5/

b) The speed of the motor $\Rightarrow n_m = (1-s)n_{sync} = (1-0,064)(1800)$

$n_m = 1685\ r/min$. 3/

The induced torque: $\tau_{ind} = \frac{P_{conv}}{\omega_m} = \frac{23.4\,kW}{(1685\,r/min)\left(\frac{1\,min}{60\,s}\right)\left(\frac{2\pi\,rad}{1\,r}\right)}$

$\Rightarrow \tau_{ind} = 132.6\,Nm$. 5/

Alternatively $\tau_{ind} = \frac{P_{AG}}{\omega_{sync}}$

c) The output power of the motor is

$P_{out} = P_{conv} - P_{mech} = 23.4\,kW - 0,300\,kW = 23.1\,kW$ 4/

$$\tau_{load} = \frac{P_{out}}{\omega_m} = \frac{23.1\,kW}{(1685\,r/min)\left(\frac{1\,min}{60\,s}\right)\left(\frac{2\pi\,rad}{1\,r}\right)} = 130,9\,N.m$$ 5/